

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276408

Kind Code

A1

Publication Date

September 04, 2025

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FUNCTIONALLY GRADED MATERIALS FROM THERMAL PROCESSING

Abstract

An additive manufacturing system includes a first energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component, a second energy delivery device configured to deliver energy to the build surface of the component; a stage configured to support an additively-manufactured component, at least one heat sensor configured to capture data indicative of a temperature of a portion of a component, and a computing device. The computing device is configured to receive data from the at least one heat sensor; and control the first or the second energy device based at least partially on the received data from the at least one heat sensor to provide functionally-graded characteristics to the additively-manufactured component, in-situ, through modification of an amount of thermal energy delivered by the first energy delivery device or the second energy delivery device.

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Family ID: 96881623

Appl. No.: 18/593438

Filed: March 01, 2024

Publication Classification

Int. Cl.: B23K26/342 (20140101); B22F10/28 (20210101); B22F10/36 (20210101); B22F10/85 (20210101); B22F12/90 (20210101); B23K26/03 (20060101); B23K26/144 (20140101); B23K26/70 (20140101); B33Y10/00 (20150101); B33Y30/00 (20150101); B33Y50/02 (20150101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD The disclosure relates to additive manufacturing techniques.

BACKGROUND

[0001] Additive manufacturing generates three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing may utilize powdered materials and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures.

SUMMARY

[0002] In some examples, the disclosure describes an additive manufacturing system which includes a first energy delivery device configured to deliver energy to a build surface of an additively-manufactured component to form a melt pool in the build surface of the component, a second energy delivery device configured to deliver energy to the build surface of the additively-manufactured component, a powder delivery device configured to direct a powder stream toward the melt pool, at least one heat sensor configured to measure a temperature of a portion of the component, and a computing device. The computing device is configured to receive data from the at least one heat sensor captured at a first point in time and captured at a second point in time. The computing device is further configured to determine a thermal history of the component based at least partially on the received data captured at the first point in time and the received data received data captured at the second point in time, and control the first energy delivery device or the second energy delivery device based at least partially on the determined thermal history.

[0003] In some examples, the disclosure is directed to techniques which include receiving, by a computing device, data captured at a first point in time from at least one heat sensor of an additive manufacturing system. The heat sensor is configured to measure a temperature of a portion of a component being additively-manufactured. The additive manufacturing system comprises a powder delivery device configured to direct a powder stream toward melt pool in a build surface of the additively-manufactured component, a first energy delivery device configured to deliver energy to the build surface of a component to form the melt pool, and second energy delivery device configured to deliver energy to the build surface, each mass sensor associated with a portion of the additive manufacturing system. The technique includes receiving, by the computing device, data captured at a second point in time from the at least one heat sensor and determining, by the computing device, a thermal history of the portion of the component based at least partially on the received data captured at the first point in time and the received data captured at the second point in time. The technique further includes controlling, by the computing device, the first energy delivery device or the second energy delivery device based at least partially on the determined thermal history of the component.

[0004] In some examples, the disclosure describes an additive manufacturing system which includes a first energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component, a powder delivery device configured to direct a powder stream toward the melt pool, and a second energy delivery device

configured to deliver energy to the build surface of the component. The additive manufacturing system further includes a stage configured to support an additively-manufactured component, at least one heat sensor configured to capture data indicative of a temperature of a portion of a component, and one or more computing devices. The computing device or devices are configured to receive data from the plurality of heat sensors, and control the first or the second energy device based at least partially on the received data from the plurality of heat sensors to provide functionally-graded characteristics to the additively-manufactured component, in-situ, through modification of the amount of thermal energy delivered by the first energy delivery device or the second energy delivery device.

[0005] In some examples, the disclosure is directed to a technique which includes a method which includes receiving, by one or more computing devices, data from at least one heat sensor configured to measure a temperature of an additive manufacturing system, wherein the additive manufacturing system comprises a powder delivery device configured to direct a powder stream toward a melt pool in a build surface of an additively-manufactured component mechanically supported by a stage, a first energy delivery device configured to deliver energy to the build surface of a component to form the melt pool, a second energy delivery device configured to deliver energy to the build surface of the component. The technique includes controlling, by the one or more computing devices, the first or the second energy device based at least partially on the received data from the at least one heat sensor to provide functionally-graded characteristics to the additively-manufactured component, in-situ, through modification of the amount of thermal energy delivered by the first energy delivery device or the second energy delivery device.

[0006] In some examples, the disclosure describes a functionally-graded, additively-manufactured component. The functionally-graded, additively-manufactured component includes a layer-by-layer built component body formed by directing, via one or more computing devices, a powder delivery device to deliver a powder stream toward a melt pool on a build surface of the component body, wherein the component body is mechanically supported by a stage, and wherein the melt pool is formed by energy from a first energy delivery device. The component body comprises a first portion and a second portion, the first portion and the second portion differing in at least one of a strength, a hardness, a ductility, or a microstructure. Forming the first portion and the second portion comprises receiving data from at least one heat sensor indicative of a temperature of a portion of the component, and modifying the amount of energy delivered by the first energy delivery device or a second energy device based at least partially on the received data from the at least one heat sensor to form the first portion and the second portion of the component body in-situ.

[0007] In some examples, the disclosure is directed to an additive manufacturing system which includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component and a powder delivery device configured to direct a powder stream toward the melt pool. The additive manufacturing system also includes a microstructural monitoring device configured to capture data representative of a microstructure of at least a portion of the component and a computing device. The computing device is configured to receive data from the microstructural monitoring device, and control at least one of the powder delivery device or the energy delivery device based at least partially on the data received from the microstructural monitoring device.

[0008] In some examples, the disclosure is directed to a method which includes receiving, by a computing device of an additive manufacturing system, data representative of at least a portion of a component being built from a microstructural monitoring device of the additive manufacturing system. The additive manufacturing system further includes a powder delivery device configured to direct a powder stream toward a melt pool in a build surface of an additively-manufactured component and an energy delivery device configured to deliver energy to the build surface of the additively-manufactured component to form the melt pool. The method includes controlling, by the computing device, at least one of the powder delivery device or the energy device based at least

partially on the received data from the microstructural monitoring device.

[0009] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a conceptual block diagram illustrating aspects of an example additive manufacturing system that includes a powder source mass sensor, a powder flow monitoring system, a topology sensor, a microstructural monitoring device, a first energy delivery device, a second energy delivery device, and a computing device configured to control the additive manufacturing system during an additive manufacturing technique.

[0011] FIG. 2 is a conceptual and schematic diagram illustrating further aspects of the additive manufacturing system of FIG. 1.

[0012] FIG. 3 is a conceptual diagram illustrating an example of portions of a powder stream imaged by a powder flow monitoring system.

[0013] FIG. 4 is an example calibration curve of particle detections versus mass flow.

[0014] FIG. 5 is a conceptual block diagram illustrating portions of the example additive manufacturing system of FIG. 1, including an optical system for observing thermal emissions around a melt pool and a thermal camera for monitoring a size of the melt pool.

[0015] FIG. 6 is a conceptual block diagram illustrating an example optical system for observing thermal emissions around a melt pool formed during the additive manufacturing technique.

[0016] FIG. 7 is a process flow diagram illustrating a mass flux monitoring, heat flux monitoring, and microstructural monitoring and control technique.

[0017] FIG. 8 is a conceptual and schematic diagram illustrating an additively-manufactured, functionally-graded component.

[0018] FIG. 9 is a flow diagram illustrating an example technique according to the present disclosure.

[0019] FIG. 10 is a flow diagram illustrating an example technique according to the present disclosure.

[0020] FIG. 11 is a flow diagram illustrating an example technique according to the present disclosure.

DETAILED DESCRIPTION

[0021] The disclosure generally describes techniques and systems for monitoring mass flux and heat flux in a blown powder additive manufacturing technique, such as a directed energy deposition (DED) technique. During blown powder additive manufacturing, a component is built up by adding material to the component in sequential layers. The final component is composed of a plurality of layers of material. In some blown powder additive manufacturing techniques for forming components from metals or alloys, an energy source may direct energy at a substrate to form a melt pool. A powder delivery device may deliver a powder to the melt pool, where at least some of the powder at least partially melts and is joined to the melt pool and, thus, the substrate.

[0022] The properties of the final component, including the presence or absence of material defects and the resulting microstructure, are a function of a number of variables related to mass flux and heat flux. As such, measurement, modeling, control, and validation of mass flux and heat flux within the blown powder additive manufacturing system may enable characterization or prediction of final component properties, control of the blown powder additive manufacturing technique during the process, quality assurance for the final component, development of new blown powder additive manufacturing techniques, and the like.

[0023] Several challenges may arise while performing additive manufacturing techniques with additive manufacturing systems. For example, in some cases additively-manufactured components may require complex heat treatments post-deposition (e.g., after the component has been manufactured to substantially the final shape) to achieve required properties. Post deposition heat treatments may be costly and time consuming to perform, and may add additional steps to a manufacturing process. Furthermore, such post-deposition heat treatments may be difficult to perform locally, such as on a repaired component, because heat treatments may lead to deformation of non-repaired areas. Example components may include, but are not limited to, for example, seal segments, shrouds, combustion tubes, blade tracks, disc assemblies, airfoils (e.g., blades or vanes), combustion chamber liners, or the like, of a gas turbine engine.

[0024] Where the additively-manufactured component is a compressor blade, as one example, different material properties may be desirable at different portions of the blade. For example, different portions of the blade may include different stresses along the length during operation of a gas turbine engine. High ductility may be desirable at radially outer (relative to an engine centerline) portions of the blade, while high hardness may be desirable at a leading edge. It may be difficult to impart desired properties to specific areas of the blade as a post-process. It should be noted that turbine blades are not the only components where different material properties, attainable by heat-treatment or controlled cooling, are desirable, and the disclosed systems and techniques may be useful for additively manufacturing such components in-situ. For example, any rotative component or component that requires a harder surface to resist wear but also needs ductility elsewhere (e.g., gears, transmission components, ablative protective panels, sealing surfaces, or the like) may also be manufactured according to aspects described herein.

[0025] In accordance with one or more aspects this disclosure, an additive manufacturing system may be configured to thermally monitor and selectively tailor heat treatment to an additively-manufactured component in-situ (e.g., while the component is being manufactured or before removal of the component from a stage on which the component was additively-manufactured). In this way, rather than requiring post-deposition heat treatment, tailored properties may be delivered to one or more portions of the component while it is being manufactured, which may remove processing steps and/or allow for functional gradation of the component. To perform the thermal monitoring and selective in-situ heat treatment, the additive manufacturing system may include at least one heat sensor configured to capture data indicative of a portion of the component. In some examples, each portion of a component being additively-manufactured may be associated with a corresponding heat sensor configured to monitor the temperature of the portion.

[0026] The heat sensor or sensors may be configured to capture data over time (e.g., at a first point in time, a second point in time, a third point in time, etc.), and the captured data may be stored and analyzed by a computing device of the additive manufacturing system. The captured data may correspond to a portion of the additively-manufactured component and/or the system during some or all of the additive manufacturing process. Thus, the captured data from the heat sensor(s) may be used to determine a thermal history of the component (e.g., by comparing the data captured at the first point in time to the data captured at the second point in time). The determined thermal history may be compared to a model, material properties, or the like to determine or predict material properties (e.g., a strength, a hardness, a ductility, a porosity, a microstructure, a cracking behavior, or the like) of the portion of the component. The computing device or devices may receive and store data from additional heat sensors, each corresponding to another portion of the component. In this way, in some examples, a matrix may be generated having the thermal history of different portions of the additively-manufactured component. The computing device may modify energy delivered to the build surface by at least one energy delivery device (e.g., an on-axis laser, an off-axis laser, an induction heater, a microwave heater, a cooling device, or the like) to better align the portion of the component to the desired thermal history. For example, the second energy delivery device may be configured to deliver energy to the portion of the component such that the

temperature of the portion is maintained at a desired temperature for a certain length of time, which may correspond to a desired tempering, densification, or the like of the portion of the component. In this way, the additive manufacturing system may be configured to selectively tailor the energy delivered to build a component with the desired properties.

[0027] Some components may experience various stress states during service. For example, aerospace components such as those mentioned above may require different properties in different portions of the component due to the expected stresses in the different portions. Functionally-graded components (e.g., differing in strength, ductility, porosity, microstructure, or the like to different portions of the component may be difficult to create through conventional manufacturing techniques, and/or may require one or more post-processes (e.g., heat treatment) to modify (e.g., temper) selected portions of the component or otherwise impart desired characteristics to particular areas.

[0028] In accordance with one or more aspects of the present disclosure, functionally-graded components may be additively-manufactured in-situ using systems and techniques capable of imparting desired material properties at selected portions of the component. The disclosed systems and techniques may be used to generate advanced components capable of handling different types of stresses during operation. A functionally-graded, additively-manufactured component may be a component formed through a layer-by-layer deposition process. Modification of energy delivered to different portions of the component by a first energy delivery device or a second energy delivery device may be used to generate the functionally-graded component. Put differently, the component may be heat-treated in-situ to create portions of the component that have different properties, such as a first portion, a second portion, a third portion, or the like. The first portion, the second portion, and other portions may differ in one or more of a strength, a ductility, a porosity, a microstructure, a modulus of elasticity, or the like.

[0029] As mentioned above, conventional directed energy deposition systems or other additive manufacturing systems may require post-processing to impart different functional characteristics to different portions of the component. In accordance with one or more aspects of the current disclosure, additive manufacturing systems may include more than one energy delivery device. Inclusion of a second energy delivery device, an optional third energy delivery device, and/or a cooling device (e.g., configured to remove thermal energy from the build surface of the component) may allow for selective tailoring of the energy delivered to desired portions of the component to change the properties of the portion in-situ, thereby reducing or eliminating the need for post-process heat treatment.

[0030] In some examples, a primary energy delivery device may be a laser associated with the deposition head of the additive manufacturing system that forms a melt pool in a build surface of the component for material deposition. In some examples, the primary (or “first”) energy delivery device may be located coincident with a central longitudinal axis of the deposition head. Secondary and/or tertiary energy delivery device may be configured to add or remove energy to other localities on the build surface (e.g., by one or more off-axis lasers) or by another energy delivery device, (e.g., an induction heater, an infrared heater, gas impingement system, a microwave heater, or the like) which may be configured to deliver energy to particular localities on the component or globally to the component. In some examples, such secondary and/or tertiary energy delivery devices may be displaced from the central longitudinal axis of the deposition head, and thus be called “off-axis.” In some examples, secondary and/or tertiary energy delivery devices may be configured to preheat a portion of the component prior to formation of the melt pool in the portion by the primary energy delivery device. In some examples, secondary and/or tertiary energy delivery devices may deliver energy to the build surface simultaneously to the primary energy delivery device delivering energy to the build surface. Similarly, the secondary and/or tertiary energy delivery devices may be configured to deliver energy after (e.g., subsequent to) the primary energy delivery device forming the melt pool on the build surface. In this way, a cooling rate (e.g., a

solidification rate) of the layer after the primary energy delivery device passes may be controlled by modification of energy delivered by one or more supplemental energy delivery devices.

[0031] In accordance with one or more aspects of this disclosure, an additive manufacturing system may include a plurality of sensors for sensing mass flow at various points along the powder flow and at least one heat sensor for sensing heat flow within the system. For instance, the mass flow sensors may include a mass sensor associated with a powder source, a powder flow monitoring system sensing powder flow between an output of a powder delivery device and the melt pool, and a topology sensor for measuring a topology of material added to the melt pool. The at least one heat sensor may include at least one sensor for monitoring a size and/or temperature of the melt pool, and at least one heat sensor (e.g., temperature probe) for monitoring a heat flow (e.g., cooling rate or a sonification rate) around the melt pool. The sensors may output data to a computing device, which may analyze and/or controls the additive manufacturing system based on the received sensor data. By monitoring mass flow at different points along the powder flow and monitoring heat flow in multiple ways, the system described herein may enable a more complete understanding of mass and heat flux within the system. Additionally, in some examples, the computing device may control the first or the second energy device based at least partially on the received data from the at least one heat sensor to provide functionally-graded characteristics to the additively-manufactured component, in-situ, through modification of the amount of thermal energy delivered by the first energy delivery device or the second energy delivery device.

[0032] Additively-manufactured components may be subject to quality risks during manufacture. For examples, contaminants or agglomerations in powder in a stream of powder may be delivered to the melt pool and incorporated into the component, or splatter may evacuate the melt pool and adhere improperly to the build surface, or the like. In conventional systems, destructive testing and/or extensive quality assurance testing after manufacturing the component may be required to ensure that the component meets microstructural quality standards. Where destructive testing is employed, complete quality assurance may not be ensured because the microstructure may vary during a single build or deposit.

[0033] In accordance with one or more examples of the current disclosure, additive manufacturing systems and techniques may incorporate one or more microstructural monitoring devices configured to capture data representative of the component's microstructure in-situ (e.g., during the build). The microstructural monitoring device may include, for example, an X-Ray device, a computed tomography (CT) device, a magnetic resonance imaging (MRI) device, an ultrasound device, another acoustic monitoring device, or the like. Accordingly, the data captured by the microstructural monitoring may be representative of a layer or portion of the component being additively manufactured, and the data may be output by the microstructural monitoring device to the computing device for analysis and/or control of the additive manufacturing system. For example, a modulus of elasticity, a microstructural texture, a porosity, and/or a cracking behavior of the component may be determined by the computing device based at least partially on the data captured by the microstructural monitoring device. Based at least partially on the data captured by the microstructural monitoring device, the computing device may modify one or more of the power delivered to the energy delivery device(s), the travel speed of the deposition head, the spot size of the primary energy delivery device, the power density, shielding gas flow rates, powder delivery gas flow rate, the build strategy, or other parameters to control the microstructure of the component while the component is being built. Furthermore, the computing device may be configured to stop the build, or output a representation of a microstructure failing to meet a threshold setting, if an anomaly or problem is detected (e.g., the data captured by the microstructural does not conform to the expected values). The problem may be addressed before the build is continued, which may save time and repair costs. In this way, the microstructure of the component may be determined in-situ and used to inform and/or control a build strategy. In some examples, the microstructural monitoring device or computing device may house a machine learning algorithm, and the machine

learning algorithm may be trained on data generated by the microstructural monitoring device. For example, the machine learning algorithm may be used to correlate the data generated by the microstructural monitoring device to data indicative of materials quality of powder in the powder stream. In this way, in-situ microstructural monitoring of an additively-manufactured component may be used to validate, modify, or stop an additive manufacturing process.

[0034] FIG. 1 is a conceptual block diagram illustrating aspects of an example additive manufacturing system 10. Additive manufacturing system 10 includes several components configured to monitor mass flow, and several components configured to monitor energy (e.g., heat flux) within system 10. System 10 includes a powder source mass sensor 44, a powder flow monitoring system (PFMS) 18, and a topology sensor 48. These components are configured to monitor mass flow of powder within additive manufacturing system 10 during an additive manufacturing technique. In the example illustrated in FIG. 1, additive manufacturing system 10 further includes a computing device 12, a powder delivery device 14, a first or primary energy delivery device 16, a second or secondary energy delivery device 17, a stage 20, a powder source 42, powder source mass sensor 44, topology sensor 48, and microstructural monitoring device (MMD) 19. Computing device 12 is operably connected to powder delivery device 14, energy delivery devices 16 and 17, PFMS 18, stage 20, powder source 42, powder source mass sensor 44, topology sensor 48, and MMD 19. FIG. 1 thus illustrates mass flow monitoring and other aspects of example additive manufacturing system 10. To simplify illustration of FIG. 1 and improve clarity of the figure, further aspects of additive manufacturing system 10 are shown in FIG. 5 and described below with reference to FIG. 5, which is more directed toward heat flow aspects of system 10.

[0035] Stage 20 is configured to mechanically support a component 22 during an additive manufacturing technique. Component 22 may be considered in-situ when mechanically supported by stage 20. In some examples, stage 20 is movable relative to energy delivery device 16 and/or energy delivery device 16 is movable relative to stage 20. Similarly, stage 20 may be movable relative to powder delivery device 14 and/or powder delivery device 14 may be movable relative to stage 20. For example, stage 20 may be translatable and/or rotatable along at least one axis to position component 22 relative to energy delivery device 16 and/or powder delivery device 14. Similarly, energy delivery device 16 and/or powder delivery device 14 may be translatable and/or rotatable along at least one axis to position energy delivery device 16 and/or powder delivery device 14, respectively, relative to component 22. Stage 20 may be configured to selectively position and restrain component 22 in place relative to stage 20 during manufacturing of component 22.

[0036] Powder source 42 is the source of powder for powder stream 30. Powder source 42 may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source 42 also may include mechanism for entraining the powder in a gas flow. For instance, powder source 42 may be coupled to a gas source, which provides a gas flowing through powder source 42 and entraining powder within the gas flow. Additionally, or alternatively, powder source 42 may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream.

[0037] System 10 may include a powder source mass sensor 44 associated with powder source 42. Powder source mass sensor 44 may be configured to quantify loss of mass in the powder source 42 or, alternatively, a mass flow out of powder source 42.

[0038] Powder source 42 is fluidically coupled to powder delivery device 14 via a flow path 46. Flow path 46 may include any suitable structure(s) defining an enclosed flow between powder source 42 and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. 1, for at least part of flow path 46 between powder source 42 and nozzles of powder delivery device 14, flow path 46 may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. 1, in some examples, flow path 46 may include one or more

nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0039] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder. The combined powder defines powder stream **30**. In some examples, powder delivery device **14** includes a single nozzle, which may be point nozzle, or a single nozzle that is an annular channel. In other examples, powder delivery device **14** includes a plurality of nozzles (e.g., three nozzles or four nozzles). Regardless of the number of nozzles, powder delivery device **14** may output a powder stream that is focused at a focus plane. As powder delivery device **14** is movable in the z-axis shown in FIG. **1** relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with build surface **28**.

[0040] At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**. At least some of the powder that impacts melt pool **32** may be joined to component **22**. In some examples, powder delivery device **14** may be mechanically coupled or attached to primary energy delivery device **16** to facilitate delivery of powder stream **30** and energy **34** for forming melt pool **32** to substantially the same location adjacent to component **22**.

[0041] Primary energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0042] In some examples, primary energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**. Primary energy delivery device may be configured to focus energy **34** from the energy source on a local spot on build surface **28** to generate melt pool **32**.

[0043] In some examples, at least a portion of primary energy delivery device **16** and powder delivery device **14** may be combined or attached to each other. For example, a deposition head (e.g., deposition head **54** of FIG. **2**) may include part of powder delivery device **14** (e.g., internal channels and powder nozzle(s) **56** for forming powder stream **30** and directing powder stream **30** toward build surface **28**) and part of primary energy delivery device **16** (e.g., the energy delivery head). As shown in FIG. **1**, in some examples, primary energy delivery device **16** may be arranged of configured such that energy **34** and powder stream **30** both exit from a common deposition head (**54**, FIG. **2**) and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel (e.g., formed along central longitudinal axis L, FIG. **2**) within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder nozzle(s) **56** for forming powder stream **30** and directing powder stream **30** toward build surface **28**. Such an arrangement between primary energy source **16** and powder delivery device **14** may be called an “on-axis” arrangement of primary energy source **16**, because both energy and powder may be delivered coaxially with a central longitudinal (Z-direction) axis of the deposition head.

[0044] System **10** also includes powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an

illumination device and an imaging device. In some examples, the illumination device may include one or more light source. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28** (e.g., central longitudinal axis **L**).

[0045] The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such greater than 1000 Hz. Because of the velocity of the powder in powder stream **30**, even such a frame rate may image only a fraction of the powder flowing between powder delivery device **14** and build surface **28**.

[0046] In some examples, PFMS **18** also includes a housing configured to enclose the illumination device and the imaging device. The housing may be configured to protect the illumination device and the imaging device from damage due to the harsh conditions to which PFMS **18** may be exposed during use. For example, the housing may protect the illumination device and the imaging device from powder deflections from powder stream **30** off build surface **28**, may cool the illumination device and the imaging device to remove heat incident on PFMS **18** from melt pool **32** and energy delivery device **16**, or the like.

[0047] PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16**, e.g., in the x-y plane shown in FIG. **1**. This may help maintain a relative x-y position of PFMS **18** and the image plane of the imaging device relative to powder stream **30**. This may facilitate analysis of image data captured by the imaging device.

[0048] PFMS **18** may be movable in the z-axis direction of FIG. **1** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may enable movement of image plane **38** along the z-axis of FIG. **1** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may allow PFMS **18** to image powder stream **30** at different positions between powder delivery device **14** and build surface **28**. In this way, PFMS **18** may analyze powder stream **30** along powder stream **30** to help determine parameters of powder stream **30** along its length.

[0049] In some example, PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16** and movable parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28** by an adjustable z-stage **40**. Adjustable z-stage **40** may be attached to energy delivery device **16**, powder delivery device **14**, or a portion of system **10** that moves energy delivery device **16** and/or powder delivery device **14**, such that PFMS **18** moves in the x-y axis in registration with energy delivery device **16** and/or powder delivery device **14**.

[0050] Adjustable z-stage **40** may be controlled by computing device **12** to position PFMS **18** and image plane **38** relative to powder stream **30**. Further, computing device **12** may control adjustable z-stage **40** to move PFMS **18** vertically and out of the way to allow powder delivery device **16** and energy delivery device **16** access to physically constrained areas, e.g., between vanes of a doublet or triplet of a nozzle guide vane for a gas turbine engine.

[0051] System **10** further includes a topology sensor **48**. Topology sensor **48** is configured to monitor an amount of powder captured by melt pool **32** by imaging melt pool **32** and the added material, allowing the mass to be quantified (e.g., by computing device **12**) using the dimensions of the added material and density of the material (powder). In some examples, topology sensor **48** includes a laser and a sensor (e.g., an imaging device), which senses laser light reflected by the structure being imaged (e.g., melt pool **32** and the added material). The laser may have a defined wavelength, which may affect the resolution of the topology sensor **48**. In some examples, the wavelength and sensor may be selected such that the resolution of topology sensor **48** is as great as about 10 microns (e.g., about 6 microns).

[0052] In some examples, topology sensor **48** may be positioned substantially directly above

component **22** and may include an interferometer, which provides depth information based on the time from outputting a laser pulse to the sensing of the reflected light. In other examples, topology sensor **48** may be positioned at an offset with respect to component **22** such that the sensor senses depth information without using an interferometer.

[0053] In some examples, topology sensor **48** may be integral with system **10**, e.g., disposed within the enclosure or working area of system **10**. In other examples, topology sensor **48** may be an add-on component to system **10**. For example, the enclosure in which the additive manufacturing technique is performed may include a transparent window, and topology sensor **48** may be positioned outside of the enclosure and may image component **22** through the transparent window.

[0054] Although a topology sensor **48** is described in the examples of this disclosure, in other examples, another metrology device may be utilized to determine the amount of powder captured by melt pool **32**. For example, another type of light source may be used. In some examples, if another type of light source is used, component **22** or stage **20** may include one or more features that serve as indicators of scale. Furthermore, although described as a single topology sensor **48**, more than one sensor may be used, and may employ more than one of the technologies described above.

[0055] System **10** further includes secondary energy delivery device **17**. In some examples, secondary energy device **17** may include an energy source and energy delivery head as described above with respect to primary energy delivery device **16**. As such, secondary energy delivery device **17** may be configured to deliver energy **34** to a second spot on build surface **28**, as will be further described below. In some examples, secondary energy delivery device **17** may be displaced from a central longitudinal axis of the deposition head, and thus may be called an “off-axis” energy source. Alternatively, secondary energy delivery device may be configured to deliver energy globally to component **22** (e.g., the entire component body **22**). Furthermore, although it is considered that the energy source for secondary energy delivery device **17** may be a laser, other sources of energy are considered. For example, the energy source for secondary energy delivery device **17** may include one or more of an induction heater, an infrared heater, a microwave heater, a fan or blower system configured to deliver hot gases to build surface **28**, or the like.

[0056] System **10** includes microstructure monitoring device (MMD) **19**. MMD **19** may be configured to capture data representative of the microstructure of component **22** in-situ, and output the captured data to computing device **12**. MMD **19** may include an imaging sensor such as an X-Ray device, a computed tomography (CT) device, a magnetic resonance imaging (MRI) device, or the like. Additionally, or alternatively, MMD **19** may include an acoustic sensing system such as an ultrasound device. Although illustrated as an off-axis add-on to system **10**, in some examples MMD **19** may be part of a deposition head and be arranged on-axis.

[0057] Computing device **12** is configured to control components of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** is configured to control operation of system **10**, including, for example, powder delivery device **14**, primary energy delivery device **16**, secondary energy delivery device **17**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, topology sensor **48**, and/or MMD **19**. Computing device **12** may be communicatively coupled to powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** using respective communication connections. In some examples, the communication connections may include network links, such as Ethernet, ATM, or other network connections. Such connections may be wireless and/or wired connections. In other examples, the communication connections may include other types of device connections, such as USB, IEEE 1394, or the like.

[0058] Although FIG. **1** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**. In general, control and

processing functions described herein may be divided among one or more computing devices. For instance, system **10** may include a controller for energy delivery devices **16** and **17**, powder delivery device **14**, and stage **20**, a separate controller for PFMS **18**, and a separate computing device for analyzing data obtained by PFMS **18**, mass sensor **44**, topology sensor **48**, and MMD **19**. As another example, system may include a dedicated controller for each of primary energy delivery device **16**, secondary energy delivery device **17**, powder delivery device **14**, stage **20**, PFMS **18**, topology sensor **48**, and MMD **19**, and a separate computing device for coordinating control of powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** and analyzing data obtained by PFMS **18** powder source mass sensor **44**, and/or topology sensor **48**. Other examples of computing system architectures for controlling system **10** and analyzing data obtained from system **10** will be apparent and are within the scope of this disclosure.

[0059] Computing device **12** may be configured to control operation of powder delivery device **14**, primary energy delivery device **16**, secondary energy delivery device **17**, adjustable z-stage **40**, stage **20**, and/or topology sensor **48** to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, topology sensor **48**, and MMD **19**. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, primary energy delivery device **16**, secondary energy delivery device **17**, adjustable z-stage **40**, topology sensor **48**, and/or MMD **19** to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery devices **16** and **17**, PFMS **18**, topology sensor **48**, and MMD **19**. Positioning component **22** relative to powder delivery device **14**, energy delivery devices **16** and **17**, PFMS **18**, topology sensor **48**, and MMD **19** may include positioning a predetermined surface (e.g., a surface to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, topology sensor **48**, and/or MMD **19**.

[0060] Computing device **12** may be configured to control system **10** to deposit layers **24** and **26** to form component body **25** and eventually finished component **22**. As shown in FIG. **1**, component **22** may include a first layer **24** and a second layer **26**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including first layer **24** and second layer **26**, the technique may be extended to components **22** with more complex geometry and any number of layers. Furthermore, although component **22** is illustrated as being uniform, in some examples component **22** may be functionally-graded and include at least two different portions having different selectively tailored properties, as will be further illustrated and described below.

[0061] To form component **22**, computing device **12** may control powder delivery device **14**, primary energy delivery device **16**, and secondary energy delivery device **17** to form, on a surface **28** of first layer of material **24**, a second layer of material **26** using an additive manufacturing technique. Computing device **12** may control primary energy delivery device **16** to deliver energy **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**. Computing device **12** may control powder delivery device **14** and primary energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** may control secondary energy delivery device **17** to deliver energy to a portion of build surface **28** to modify the cooling rate, solidification behavior, or other aspects of the thermal profile of component **22** determined based on data from a plurality of heat sensors

(not illustrated in FIG. 1 for clarity) associated with portions of component 22. The plurality of heat sensors may be configured to capture and store data during the build (e.g., at a first point in time, a second point in time, a third point in time, etc.). Computing device 12 may determine, based on the captured data by the plurality of heat sensors, the thermal history of component 22 and compare the thermal history to a model or perform a calculation to correlate the thermal history of component 22 relative to the thermal history of a component with desired characteristics or properties (e.g., strength, ductility, porosity, microstructure, or the like). Computing device 12 may then modify one or more of the magnitude, power density, travel speed, or spot size (i.e., focus area) of primary energy delivery device 16 or secondary energy delivery device 17 to generate a portion of component 22 (e.g., all or a portion of layer 26) to selectively tailor the characteristics of the portion based on the thermal history of the portion. For example, computing device 12 may cause second energy delivery device 17 to pause, dwell, or skip certain portions of build surface 28 to generate a desired microstructure. Computing device 12 may cause MMD 19 to capture data representative of a microstructure of layer 26. Computing device 12 may then analyze the captured microstructure data to determine whether the captured microstructure data meets a threshold (e.g., a threshold density, a threshold porosity, a threshold uniformity, or the like). In this way, functionally-graded, additively-manufactured component 22 may be generated by system 10 in-situ, and the

[0062] Computing device 12 then may control a z-axis position of stage 20 and/or powder delivery device 14 and primary energy delivery device 16 such that melt pool 32 will be formed on surface 36 of second layer 26, and may control powder delivery device 14 and energy delivery device 16 to move energy 34 and powder stream 30 along build surface 28 in a pattern until layer 26 is complete. Computing device 12 may control powder delivery device 14 and energy delivery device 16 similarly until all layers are formed to define a completed component 22.

[0063] In some examples, computing device 12 may store/execute one or more machine learning models which may be used to adapt control of powder delivery device 14, primary energy delivery device 16, secondary energy delivery device 17, or another component. For example, data captured by first energy delivery device 16, second energy delivery device 17, MMD 19, PFMS 18, topology sensor 48, mass source 44, or other component may be used to train the machine learning device. In some examples, the machine-learning algorithm may be used to correlate the data generated by MMD 19 with data indicative of materials quality of powder in the powder stream.

[0064] FIG. 2 is a conceptual and schematic diagram illustrating example additive manufacturing system 100. Additive manufacturing system 100 of FIG. 2 may be an example of additive manufacturing system 10 of FIG. 1. System 100 includes an example powder flow monitoring system 50 configured to monitor powder flow between a powder delivery device 52 and a build surface (not shown in FIG. 2) during an additive manufacturing technique. Powder delivery device 52 may be an example of powder delivery device 14 of FIG. 1, and PFMS may be an example of PFMS 18 of FIG. 1.

[0065] Powder delivery device 52 includes a deposition head 54 that carries a plurality of powder nozzles 56. Plurality of powder nozzles 56 output a powder stream 58 toward the build surface. As shown in FIG. 2, the powder stream 58 may be focused at a focal plane, such that powder stream 58 is converging toward the focal plane and diverging away from the focal plane. As discussed above, deposition head 54 may further include a primary energy delivery device, which is not illustrated in FIG. 2 for improved clarity.

[0066] PFMS 18 includes a housing 60 (also referred to as an enclosure), which encloses an imaging device 62 and an illumination device 64. In some examples, imaging device 62 may be a high-speed camera and illumination device 64 may be laser illuminator. Housing 60 is attached to an adjustable z-stage 66 by a bracket 68.

[0067] Housing 60 is configured to enclose imaging device 62 and illumination device 64 and help protect imaging device 62 and illumination device 64 from a surrounding environment. For

instance, housing **60** may be configured to surround imaging device **62** and illumination device **64** and prevent any powder that reflects from the build surface toward PFMS **18** from impacting imaging device **62** or illumination device **64**.

[0068] Further, housing **60** may be configured to cool imaging device **62** and illumination device **64**. Imaging device **62** and illumination device **64** may be exposed to heat from the melt pool at the build surface and energy from the energy delivery device. Imaging device **62** and illumination device **64** may be relatively sensitive to heat and have improved operational lifetime if maintained and operated below a certain temperature. PFMS **50** may include a cooling system **70** configured to remove heat from within housing **60** to cooling imaging device **62** and illumination device **64**. For instance, cooling system **70** may include cooling fluid circuit through which a cooling fluid flows, and housing **60** may include part of the cooling circuit. In some examples, housing **60** may be formed from a material having relatively high thermal conductivity, such as aluminum, to help transfer heat from within housing **60** to cooling system **70** (e.g., a cooling fluid flowing through cooling system **70**).

[0069] As described above, PFMS **50** may be configured to measure powder flow of powder stream **58** (FIG. 2) at one or more axial (or longitudinal) locations of powder stream **58** and determine one or more parameters associated with the powder flow. For instance, illumination device **64** may illuminate powder of powder stream **58** in a plane oriented substantially orthogonal to a longitudinal axis that extends from powder delivery device **52** to the build surface. PFMS **50** may be positioned at a selected axial or longitudinal location to image a selected axial or longitudinal position between powder delivery device **52** and the build surface. Imaging device **62** may be configured to image at least some of the illuminated powder.

[0070] Included in the illustration of system **100** of FIG. 2 are secondary energy delivery device **74**, MMD **78**, and tertiary energy delivery device **76**. Secondary energy delivery device **74** may be a laser configured to deliver energy locally to a portion of the additively-manufactured component. Tertiary energy delivery device **76**, in the illustrated example, is an induction heater configured to add thermal energy to the entire component. It is also considered that tertiary energy delivery **76** could be another type of heater, or even be a cooling device configured to remove thermal energy from the build surface of the additively-manufactured component. In such examples, tertiary energy delivery device **76** may include a heat exchanger. System **100** includes MMD **78**. In the illustrated example of FIG. 2, MMD **78** includes an ultrasound probe configured to direct acoustic energy **79** toward the additively manufactured component and receive acoustic energy reflected back from the additively-manufactured component for detection and analysis. Each of secondary energy delivery device **74**, MMD **78**, and tertiary energy delivery device **76** may be communicatively coupled to and under control of a computing device (**12**, FIG. 1).

[0071] FIG. 3 is a conceptual diagram illustrating an example of portions of a powder stream imaged by a powder flow monitoring system. As shown in FIG. 3, since powder is flowing in powder stream **58** at a relatively high velocity, imaging device **62** may not capture images of all the powder in powder stream **58**. The fraction of powder that imaging device **62** captures images of may be a function of average powder velocity at the image plane and a frame rate or capture speed of imaging device **62**. This is represented in FIG. 3 as “sampled” particles and “missed population” particles. The fraction of particles imaged by imaging device **62** may, in some examples, be less than about 50%, less than about 40%, less than about 30%, less than about 25%, less than about 20%, or less than about 15%.

[0072] PFMS **50** may include a computing device (e.g., computing device **12** of FIG. 1) configured to analyze images captured by imaging device **62** to identify a number of particle detections in each captured image and, optionally, derive further parameters from the number of particle detections. As such, computing device **12** may be configured to receive image data representing an image captured by imaging device **62**. The image data may include representations of illuminated powder of powder stream **58**, as imaged by imaging device **62** (e.g., as captured in an image frame by

imaging device **62**). Computing device **12** may be configured to generate a representation of powder stream based on the image data and output the representation of the powder stream for display at a display device.

[0073] For instance, computing device **12** may be configured determine a powder mass flow represented by the image data. To do so, computing device **12** may be configured to identify a number of powder particles within each image frame. In some examples, computing device **12** additionally may be configured to identify a size and/or shape of each powder particle within each image frame. Computing device **12** may be configured to implement any suitable image analysis technique to identify powder particles, and, optionally, size and/or shape of powder particles.

[0074] Once computing device **12** has identified a number of powder particles within an image frame, computing device **12** may be configured to determine a mass flow based on the number of powder particles. For example, computing device **12** may be configured to determine the mass flow based on a calibration equation or calibration curve. FIG. 5 is an example calibration curve of particle detections versus mass flow. As shown in FIG. 4, the relationship between particle detections may be substantially linear.

[0075] The relationship between particle detections and mass flow may be determined experimentally. For instance, the relationship between particle detections and mass flow may be determined for each powder type (e.g., composition, size distribution, or both), as each powder type may have a different relationship between particle detections and mass flow. The relationship may be determined experimentally by flowing a known mass of powder at a known rate, and imaging the powder. By doing this multiple times at multiple rates, the calibration curve may be generated. The curve, in the form of an equation, a look-up table, or the like, may be stored in computing device **12**, and computing device **12** may use the calibration curve to determine mass flow of a similar type of powder at a different flow rate based on particle detections.

[0076] In some examples, computing device **12** may receive image data representative of a sequence of images of illuminated powder in powder stream **58**. Each image may be associated with a time. As such, computing device **12** may select one or more images of the sequence of images and analyze the one or more images. For each selected image, computing device **12** may be configured to identify a number of particle detections and, optionally, determine a mass flow associated with powder stream **58** for each image frame.

[0077] As described above, system **10** may include both mass flow monitoring and heat flow monitoring. FIG. 1 best illustrates the mass flow monitoring aspects of system **10**. FIG. 5 is a conceptual block diagram illustrating further aspects of system **10**, best illustrating the heat flow monitoring aspects of the example system. System **10** includes an optical system **80** for observing thermal emissions around melt pool **32** and a melt pool monitor **15** including a thermal camera for monitoring a size and/or temperature of melt pool **32**. Identical reference numerals in FIGS. 1 and 5 refer to the same parts. Further, those common parts are the same or substantially identical, aside from any differences described herein.

[0078] As shown in FIG. 5, primary energy delivery device **16** includes an optical system **80**. Although optical system **80** is shown and described only as associated with primary energy delivery device **16**, secondary energy delivery device **17** may include a corresponding optical system. During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Primary energy delivery device **16** may direct energy **34** at portion **P1** of first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). The temperature and cooling rate of melt pool **32** and the surrounding areas of first layer **24** affect the microstructure of the component **22** formed using the additive manufacturing technique.

[0079] In some examples, temperature probe **21** may measure a temperature of build surface **28** in portion **P2** and output captured temperature information to computing device **12**. Although described and illustrated as a temperature probe herein, element **21** may be more generally referred to as heat sensor **21**, and is not necessarily be a temperature probe. For example, heat sensor **21** may be configured to sense heat via an optical system, as described elsewhere herein.

[0080] Different portions of component **22** may, in some examples, be defined relative to melt pool **32**. In such examples, temperature probe **21** may be movable relative to build surface **28** to maintain a spatial relationship to melt pool **32**. In such examples, first portion **P1** may be defined as encompassing the portion of build surface **28** that defines melt pool **32** (and/or melt pool **32** plus the area within 1.5 radii of melt pool **32**, 2.0 radii of melt pool **32**, or 3.0 radii of melt pool **32**). In such examples. Second portion **P2** may be defined as encompassing an area with a radius equal to melt pool **32** but trailing melt pool **32** along build surface **28** relative to a toolpath by, for example, two, four, or six radii of melt pool **32**. Other portions (**P3**, **P4**, etc.) may be defined leading melt pool **32**, orthogonal to melt pool **32**, etc., and each portion may have a corresponding temperature probe **21**.

[0081] Alternatively, portions (**P1**, **P2**, etc.) may be defined relative to the expected final dimensions of additively-manufactured component **22**. In such examples, probe **21** may be stationary or may be configured to vertically adjust as layers are added to measure a temperature of build surface **28** in same or similar location. Primary energy delivery device **16** may travel through portion **P2**, forming melt pool **32** as it travels along a toolpath. In such examples, computing device **12** may store instructions for a desired thermal history of each portion (**P1**, **P2**, etc.) and modify energy delivered by secondary energy delivery device **17** or a tertiary energy delivery device to conform to the desired thermal history of the component.

[0082] Computing device **12** may control secondary energy delivery device **17** to modify the temperature and cooling rate of portion **P2** of the build surface **28** by adding energy **34** to portion **P2** to ensure that component **P2** cools according to the desired thermal history profile, such that resulting component **22** has the desired material properties. In some examples portions **P1** and **P2** may receive different amounts of thermal energy by primary energy source **16**, secondary energy source **17**, or both. As such, portion **P1** and portion **P2** may have a different thermal history, and therefore different resulting properties. For example, portions **P1** and **P2** may differ in strength, ductility, porosity, microstructure, or the like. Although portions **P1** and **P2** are described and illustrated as displaced from each other in the X-Y plane, it should also be considered that layer **24** and **26** may be treated, in-situ, by different magnitudes or durations of energy **34** or time periods by first energy delivery device **16** and/or second energy delivery device **17**. In such examples, portions **P1** and **P2** may be displaced from each other in the Z-direction. It should be noted that computing device **12** may control primary energy delivery device and secondary energy delivery device **17** independently based on data from MP monitor **15**, optical system **80**, probe **21**, or combinations thereof. For example, computing device **12** may cause secondary energy delivery device **17** to deliver energy **34** to build surface **28** before, during, and/or subsequent to primary energy delivery device **16** delivering energy **34**. Although only portions **P1** and **P2** are illustrated in FIG. 5, it is considered that additional third, fourth, and fifth portions of component **22** may be similarly monitored and controlled. In some examples, each portion may have a corresponding probe **21** and/or a corresponding energy delivery device.

[0083] In many cases, energy **34** output by primary energy delivery device **16** and/or secondary energy delivery device **17** is very high temperature and the intensity of its thermal emissions is significantly greater than the intensity of thermal emissions from melt pool **32**, and the surrounding areas of component **22**. Similarly, thermal emissions intensity at and near the center of melt pool **32** may be significantly greater than the intensity of thermal emissions near the edge of melt pool **32** and in areas surrounding melt pool **32**. Because of this, it may be difficult to accurately measure temperature and cooling rate of areas near the edge of melt pool **32** and in areas surrounding melt

pool **32**. This results in difficulty predicting and controlling microstructure of the additively manufactured component **22**.

[0084] Optical system **80** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **80** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range.

[0085] The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device. In some examples, at least a portion of the optical train is coaxial with the axis at which energy delivery device **16** outputs energy, and the at least a portion of the optical train may be attached to or otherwise configured to move with the portion of energy delivery device **16** that directs or focuses energy **34** at or near the surface of component **22**. In this way, optical system **80** may move with energy delivery device **16** and track melt pool **32** as melt pool **32** moves across component **22**, without needing to correct for any offsets between energy delivery device **16** and optical system **80** and/or needing to correct for geometry of component **22**. In other examples, the optical train may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** may be configured to compensate for the offset and any affects this may have on the imaging, including shadowing, interference, geometry of component **22**, or the like.

[0086] Optical system **80** may include an occulting device. The occulting device is configured to reduce or block emissions (e.g., thermal emissions) that originate from the energy output by energy delivery device **16** and/or near a center of melt pool **32**, which otherwise obfuscate emissions from solidifying regions of material at or near the edge of melt pool **32** and outside of melt pool **32**. The occulting device may be a rigid occulting device or a dynamic occulting device. A rigid occulting device reduces or blocks emissions from a fixed region, e.g., from the energy **34** output by energy delivery device **16**. For instance, a rigid occulting device may include a device with fixed dimensions that is opaque to wavelengths of interest. As another example, a rigid occulting device may include an apodizing lens in which a center of the lens is substantially opaque to wavelengths of interest and opacity decreases as a function of radius.

[0087] A dynamic occulting device is configured to be controlled to occult different regions, e.g., different sizes and/or shapes. A dynamic occulting device may include a rigid occulting device that is mounted to a device that can translate the rigid occulting device along and/or perpendicularly to the optical axis. As another example, a dynamic occulting device may include an opaque and viscous liquid, such as mercury, contained between two substrates. The substrates are substantially transparent to the wavelength(s) of interest. One or both of the substrates may be movable relative to the other substrate to control the distance between the substrates. By reducing the distance between the substrates, the size of the occulting region may increase. By increasing the distance between the substrates, the size of the occulting region may decrease. As a third example, a dynamic occulting device may include a digital micromirror device. Computing device **12** may be configured to control the micromirrors of the digital micromirror device to direct emissions that originate from energy **34** output by energy delivery device **16** and/or near a center of the melt pool away from the imaging device. A digital micromirror device may enable control of both the size and shape of the region of emissions that are occulted.

[0088] FIG. **6** is a conceptual block diagram illustrating an example optical system **80** for observing thermal emissions at and/or around a melt pool **32** and or a focused spot on a build surface **28** where a melt pool is not formed (e.g., by second energy delivery device **17**, FIGS. **1** and **5**) formed during an additive manufacturing technique. Optical system **80** includes an optical train that includes first imaging optics **92**, occulting device **94**, second imaging optics **96**, and imaging device **98**. Imaging device **98** may be any suitable imaging device, including, for example, a visible

light imaging device, an infrared imaging device, an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range, a two color pyrometry imaging device, or the like.

[0089] First and second imaging optics **92** and **96** may each include one or more optical devices used to direct light to imaging device **98**. For example, First and second imaging optics **92** and **96** may each include one or more refractive optical device (e.g., a lens), one or more reflective optical device (e.g., a mirror), one or more diffractive optical devices (e.g., a grating), one or more dichroic optical devices (e.g., a dichroic filter or mirror), or the like. Although two sets of imaging optics **92** and **98** are shown in FIG. 2, in other examples, system **80** may include a single set of imaging optics or more than two sets of imaging optics.

[0090] Occulting device **94** is positioned within the optical train between first imaging optics **92** and second imaging optics **96**. In other example, occulting device **94** may be positioned between imaging device **98** and imaging optics **96** or after before imaging optics **92**. In some examples, occulting device **94** is positioned as the optical component nearest imaging device. This effectively results in removal of the portion of the image which occulting device **94** blocks. In other examples, occulting device **94** is positioned at another position within the optical train **80** where the image of component **22** resolves. Imaging optics **96** then may be configured to image occulting device **94** onto imaging device **98**.

[0091] As shown in FIG. 6, in some examples, at least a portion of optical system **80** is coaxial with the axis at which energy delivery device **16** outputs energy **34** (i.e., a central longitudinal axis). For example, at least a portion of second imaging optics **92** (e.g., the portion at which emitted light **104** is incident upon second imaging optics **92**) may be coaxial with the axis at which energy delivery device **16** outputs energy **34**. This may reduce image manipulation that otherwise may be applied to the resulting image to correct for geometry of component **22**, angular offset of optical system **80** relative to energy delivery device **16**, shadowing due to the angular offset, interference, or the like. In other examples, optical system **80** (e.g., the portion at which thermal emissions **104** are incident upon second imaging optics **92**) may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** (FIG. 1) or another computing device may be configured to manipulate the resulting image to compensate for geometry of component **22**, angular offset of optical system **80** relative to energy delivery device **16**, shadowing due to the angular offset, interference, or the like.

[0092] FIG. 6 also illustrates energy delivery device **16** outputting energy **34**, which is incident upon component **22** and results in formation of melt pool **32**. Surrounding melt pool is a cooling zone **102**, in which temperature gradients from the temperature of melt pool **32** to ambient temperature are present. As shown in FIG. 6, melt pool **32** and cooling zone **102** emit thermal emissions **104** (e.g., thermal radiation), which travel through optical system **80** to imaging device **98**, which images the thermal emissions **104**. Occulting device **94** occults (e.g., reduces the intensity of or substantially eliminates) thermal emissions **104** from a selected region, e.g., a region corresponding to energy **34** and at least a portion of melt pool **32**. This may allow imaging device **98** to more effectively image relatively lower intensity thermal emissions from at or near the edge of melt pool **32** and within cooling zone **102**. This may enable more accurate measurement of temperatures within the cooling zone **102**, and heat flow within cooling zone **102**.

[0093] Returning to FIG. 5, system **10** also includes melt pool monitor (“MP monitor”) **82**. Melt pool monitor **82** may include a sensor for monitoring a characteristic of melt pool **32**. The sensor may include an imaging system, such as a visual or thermal camera, e.g., camera to visible light or infrared (IR) radiation. A visible light camera may monitor the geometry of the melt pool, e.g., a width, diameter, shape, or the like. A thermal (or IR) camera may be used to detect the size, temperature, or both of the melt pool. In some examples, a thermal camera may be used to detect the temperature of the melt pool at multiple positions within the melt pool, such as a leading edge, a center, and a trailing edge of the melt pool. In some examples, the imaging system may include a

relatively high speed camera capable of capturing image data at a rate of tens or hundreds of frames per second or more, which may facilitate real-time detection of the characteristic of the melt pool. [0094] FIG. 7 is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique. The technique of FIG. 7 may be implemented by system **10** of FIGS. 1 and 5 and will be described with concurrent reference to FIGS. 1 and 5. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. 7 may be performed by other systems. For example, system **100** of FIG. 2 may perform the described technique.

[0095] One or more computing devices **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. 7). For instance, one or more computing devices **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**.

[0096] One or more computing devices **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. Data received from powder flow mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. Data from topology sensor **48** indicates powder mass captured by melt pool **32** and added to component **22**.

[0097] One or more computing devices **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. For example, one or more computing devices **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32**, as determined based on data from topology sensor, into the mass flow determined based on data received from PFMS **18**.

[0098] Further, one or more computing devices **12** may determine an overall mass flux using the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. One or more computing devices **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. 7).

[0099] Similarly, one or more computing devices **12** may be configured to control first energy delivery device **16** and second energy delivery device **17** to deliver energy **34** to first layer **24** to establish a given heat input (see bottom left of FIG. 7). For example, one or more computing device **12** may control one or more operating parameters of energy delivery device **16** and/or second energy delivery device **17**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**). In some examples, computing device **12** may be configured to operate one of the energy delivery devices (e.g., first energy delivery device **16**) at a constant heat input and to control second energy delivery device **17** to modify (e.g., selectively tailor) properties of a first portion **P1** and/or a second portion **P2**. For example, second energy delivery device **17** may input thermal energy to portion **P2** to modify, for example, a hardness of portion **P2** of component **22**. In this way, component **22** may be thermally treated in-situ to provide functional gradation to component **22**.

[0100] One or more computing devices **12** may be configured to receive from one or more heat sensors, such as optical system **80** and/or melt pool monitor **82**. One or more computing devices may determine a cooling rate and associated heat from using data from optical system **80** and may determine a heat input into component using a size and/or temperature of melt pool **32** as observed by melt pool monitor **15**, probe **21**, or both. One or more computing devices **12** may be configured to capture data from melt pool monitor **15**, probe **21** throughout a period of time (e.g., at a first

point in time, a second point in time, a third point in time, etc.). One or more computing devices **12** may be configured to determine a thermal history of portions **P1**, **P2** of component **22** based on the captured data. One or more computing devices **12** then may use the thermal history as an input to the control algorithm used to control first energy delivery device **16**, second energy delivery device **17**, or both. (see left of FIG. 7).

[0101] In some examples, microstructural monitoring device **19** may be configured to sense, capture, and output to one or more computing device **12** data representative of a microstructure of portions **P1**, **P2** of component **22**. One or more computing devices **12** may be configured to determine a microstructure of portions **P1**, **P2** based on the captured data. One or more computing devices **12** may then use the determined microstructure as an input to the control algorithm used to control first energy delivery device **16**, second energy delivery device **17**, or both. In some examples, for example of MMD **19** meets a threshold level in one or more microstructural characteristics, one or more computing devices **12** may be configured to continue the build based on the validated microstructure of component **22**. In some examples, if one or more computing devices **12** determines that the component microstructure does not meet a threshold level, one or more computing devices may be configured to cause system **10** to stop before building a flawed component **22**, or may be configured to output a display indicating an improper microstructure.

[0102] In some examples, one or more computing devices **12** also may use the deposit topology (captured powder mass) and/or capture efficiency metric in the determination of the heat flux, as the added powder mass and quench effects associated with the captured powder affect the cooling rate.

[0103] FIG. 8 is a conceptual and schematic diagram illustrating an example additively-manufactured, functionally-graded component **222**. Component **222** includes a layer-by-layer built component body **225**, and thus includes layers **224A**, **224B**, and **224C**. Component **222** is described as being formed in-situ by system **10** of FIGS. 1 and 5, but it should be understood that system **100** of FIG. 2 may be used to form component **222**. Component **222** includes portions **P1**, **P2**, **P3**, and **P4** which are formed by modification of the amount of energy delivered to first energy source **16** or second energy source **17**. In the illustrated example of FIG. 8, each of portions **P1**, **P2**, **P3**, and **P4** each differ in at least one of a strength, a hardness, a ductility, or a microstructure from at least one of the other portions. wherein forming the first portion and the second portion comprises:

[0104] FIG. 9 is a flow diagram illustrating example technique **300** according to the present disclosure. Technique **300** is described with respect to system **10** of FIGS. 1 and 5, but technique **300** may be performed with system **100** of FIG. 2, or with other additive manufacturing systems. System **10** of FIGS. 1 and 5 and system **100** of FIG. 2 may be used to perform other techniques, such as technique **400** of FIG. 10 or technique **500** of FIG. 11.

[0105] Technique **300** includes receiving, by computing device **12**, data captured at a first point in time from temperature probe **21** representative of the temperature of portion **P2** of additively-manufactured component **22** of additive manufacturing system **10** (**302**). Technique **300** may include directing powder delivery device **14** to direct powder stream **30** toward melt pool **32** in build surface **28** of additively-manufactured component **22** under control of computing device **12** and delivering energy **34** to build surface **28** via primary energy deliver device **16** to form melt pool **32** under control of computing device **12**. In some examples, technique **300** further includes directing energy **34** from secondary energy delivery device **17** to build surface **28** to impart desired properties to portion **P2** of component **22**.

[0106] Technique **300** includes receiving, by computing device **12**, data captured at a second point in time from temperature probe **21** indicative of a temperature of portion **P2** (**304**). In some examples, technique **300** may include receiving data captured at a third point in time from temperature probe **21**, data captured at a fourth point in time from temperature probe **21**, or the like. In some examples, computing device **12** may cause a measurement of the temperature in portion **P2**

to be captured and stored at defined points in time (e.g., separated by one second, two seconds, 1 minute, 2 minutes, or the like.) Accordingly, temperature or other data indicative of heat flux in system **10** and component **22** may be captured and stored throughout some or all of an additive manufacturing process to build component **22**. The captured data may correlate to individual portions of component **22**.

[0107] Technique **300** may include analyzing the data captured by probe **21** of portion **P2**. Technique **300** may include determining, by computing device **12**, a thermal history of the component based at least partially on the received data captured at the first point in time and the received data captured at the second point in time (**306**). Technique **300** may further include controlling, by computing device **12**, primary energy delivery device **16** or secondary energy delivery device **17** based at least partially on the determined thermal history of the component (**308**). For example, computing device **12** may determine, based on the thermal history of portion **P2**, that energy **34** should be added via secondary energy delivery device **12** to portion **P2** to modify the desired cooling or solidification rate of portion **P2** to generate the desired strength, ductility, porosity, and/or microstructure of portion **P2** of component **22**. For example, computing device **12** may cause the power, travel speed, spot size, or power density of energy delivery device **17** to more closely conform to the desired thermal history of portion **P2**, based on a model, a machine learning algorithm, or the like.

[0108] FIG. **10** is a flow diagram illustrating example technique **400** according to the present disclosure. Technique **400** includes receiving, by computing device **12**, data from heat sensor **21** configured to measure a temperature of a portion **P2** of component **22** being built by additive manufacturing system **10** (**402**). Technique **400** may include directing powder delivery device **14** to direct powder stream **30** toward melt pool **32** in build surface **28** of additively-manufactured component **22** under control of computing device **12** and delivering energy **34** to build surface **28** via primary energy deliver device **16** to form melt pool **32** under control of computing device **12**.

[0109] Technique **400** includes controlling, by the one or more computing devices, the first energy delivery device **16** or second energy device **17** based at least partially on the received data from heat sensor **21** to provide functionally-graded characteristics to additively-manufactured component **22**, in-situ, through modification of the amount of thermal energy **34** delivered by first energy delivery device **16** or second energy delivery device **17** (**404**).

[0110] FIG. **11** is a flow diagram illustrating example technique **500** according to the present disclosure. Technique **500** includes receiving, by computing device **12** of system **10**, data representative of a microstructure of portion **P2** of additively-manufactured component **22** from MMD **19** (**502**). Technique **500** further includes controlling, by computing device **12**, at least one of powder delivery device **14**, primary energy device **16**, or secondary energy delivery device **17** based at least partially on the received data from microstructural MMD **19** (**504**). In some examples, computing device **12** may control first energy delivery device **16** and/or second energy delivery device **17** by modifying at least one of a power, a travel speed, a spot size, or a power density of first energy delivery device **16** and/or second energy delivery device **17**. In some examples, computing device **12** may control powder delivery device **14** by modifying at least one of a power, a travel speed, a gas flow rate, or a mass flow rate of the powder.

[0111] In some examples, technique **500** may include operating at least one of an X-Ray device, a computed tomography (CT) device, an ultrasound device, or an acoustic monitoring device as MMD **19**. In some examples, the captured data may be representative of at least one of a modulus of elasticity, a microstructural texture, or a porosity of portion **P2** of component **22**.

[0112] In some examples, computing device **12** may compare data captured by MMD **19** to a model or expected value table for an ideal additively-manufactured component **22**. Additionally, or alternatively, computing device **12** may execute a machine learning algorithm, which may be trained on data generated by microstructural monitoring device **19**. In some examples, the machine learning algorithm may be used to correlate the data generated by MMD **19** to data indicative of

materials quality of powder in the powder stream (58, FIG. 2).

[0113] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0114] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0115] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0116] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0117] Various examples have been described. These and other examples are within the scope of the following examples and claims.

[0118] Example 1: An additive manufacturing system includes a first energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a second energy delivery device configured to deliver energy to the build surface of the component; a stage configured to support an additively-manufactured component; at least one heat sensor configured to capture data indicative of a temperature of a portion of a component; and one or more computing devices configured to: receive data from the at least one heat sensor; and control the first or the second energy device based at least partially on the received data from the at least one heat sensor to provide functionally-graded characteristics to the additively-manufactured component, in-situ, through modification of an amount of thermal energy delivered by the first energy delivery device or the second energy delivery device.

[0119] Example 2: The additive manufacturing system of example 1, further comprising the additively-manufactured functionally-graded component, wherein the additively-manufactured

functionally-graded component comprises a first portion and a second portion, wherein the first portion is different from the second portion in at least one of a strength, a hardness, a microstructure or a ductility.

[0120] Example 3: The additive manufacturing system of any of examples 1 and 2, wherein the one or more computing device is further configured to: determine an overall heat flux based on the data from the at least one heat sensor; and control the first energy delivery device and the second energy delivery device based on the overall heat flux.

[0121] Example 4: The additive manufacturing system of any of examples 1 through 3, wherein: the first energy deliver device is coincident with a central longitudinal axis of a deposition head, and the second energy delivery device is not coincident with the central longitudinal axis of the deposition head.

[0122] Example 5: The additive manufacturing system of example 4, wherein: the first energy delivery device comprises a laser, and the second energy delivery device comprises a laser, an induction heater, an infrared heater, a gas impingement device, or a microwave heater.

[0123] Example 6: The additive manufacturing system of any of examples 1 through 5, further comprising a third energy delivery device.

[0124] Example 7: The additive manufacturing device of any of examples 1 through 6 wherein, to control the first energy delivery device or the second energy delivery device to provide the functionally-graded characteristics to the additively-manufactured component, the computing device is configured to control the first energy delivery device by modifying at least one of a power, a travel speed, a spot size, or a power density of the first or the second energy delivery device.

[0125] Example 8: The additive manufacturing system of any of examples 1 through 7, wherein the second energy delivery device is configured to deliver energy to the build surface of the component simultaneously with the first energy delivery device delivering energy to the build surface of the component.

[0126] Example 9: The additive manufacturing system of any of examples 1 through 8, wherein the second energy delivery device is configured to deliver energy to the build surface of the component subsequent to the first energy deliver device delivering energy to the build surface of the component.

[0127] Example 10: The additive manufacturing system of any of examples 1 through 9, wherein the one or more computing devices are configured to: determine a solidification rate of material surrounding the melt pool based on data received from an optical system, and control the second energy delivery device to modify the determined solidification rate.

[0128] Example 11: The additive manufacturing system of any of examples 1 through 10, wherein the second energy delivery device is configured to deliver energy to a local area of the build surface, wherein the local area comprises about 20 percent or less of the surface area of the build surface.

[0129] Example 12: The additive manufacturing system of any of examples 1 through 11, wherein the second energy delivery device is configured to deliver energy to a global area of the build surface.

[0130] Example 13: The additive manufacturing system of any of examples 1 through 12, further includes receive data from the plurality of mass sensors; determine an overall mass flux based on the data from the plurality of mass sensors; and control the powder delivery device based on the overall mass flux to generate the additively-manufactured component with functional characteristics.

[0131] Example 14: The additive manufacturing system of example 13, wherein the plurality of mass sensors comprises a powder flow monitoring system includes an illumination device configured to illuminate at least some powder the powder stream between the powder delivery device and the build surface; and an imaging device configured to image the illuminated powder at

an image plane that intersects an longitudinal axis of a deposition head, and wherein the one or more computing devices is configured to determine a mass flow rate of powder from the powder delivery device using data from the powder flow monitoring system.

[0132] Example 15: The additive manufacturing system of any of examples 1 through 14, further comprising a cooling device, and wherein the one or more computing devices is configured to control the cooling device to remove thermal energy from the build surface.

[0133] Example 16: The additive manufacturing system of any of examples 13 through 15, wherein the plurality of mass sensors comprise a topology sensor configured to measure a topology of material added to the melt pool, wherein the one or more computing devices is further configured to determine a mass of powder added to the melt pool based on the topology of the material added to the melt pool and a density of the powder.

[0134] Example 17: The additive manufacturing system of example 16, wherein the one or more computing devices is further configured to determine a capture efficiency by dividing the mass of powder added to the melt pool by the mass of powder leaving the powder delivery device or dividing a mass rate of powder added to the melt pool by a mass flow rate of powder leaving the powder delivery device.

[0135] Example 18: The additive manufacturing system of any of examples 14 through 17, wherein the one or more computing devices is configured to determine an overall heat flux based on the data from the at least one heat sensor and a capture efficiency.

[0136] Example 19: The additive manufacturing system of any of examples 1 through 18, wherein the computing device is configured to: determine a solidification rate of material surrounding the melt pool based on data received from an optical system., and control the second energy delivery device to modify the determined solidification rate.

[0137] Example 20: A method includes receiving, by one or more computing devices, data from at least one heat sensor configured to measure a temperature of a portion of a component of an additive manufacturing system, wherein the additive manufacturing system comprises a powder delivery device configured to direct a powder stream toward a melt pool in a build surface of an additively-manufactured component mechanically supported by a stage, a first energy delivery device configured to deliver energy to the build surface of a component to form the melt pool, a second energy delivery device configured to deliver energy to the build surface of the component; and controlling, by the one or more computing devices, the first or the second energy device based at least partially on the received data from the at least one heat sensor to provide functionally-graded characteristics to the additively-manufactured component, in-situ, through modification of an amount of thermal energy delivered by the first energy delivery device or the second energy delivery device.

[0138] Example 21: A functionally-graded, additively-manufactured component includes a layer-by-layer built component body formed by directing, via one or more computing devices, a powder deliver device to deliver a powder stream toward a melt pool on a build surface of the component body, wherein the component body is mechanically supported by a stage, and wherein the melt pool is formed by energy from a first energy delivery device, wherein the component body comprises a first portion and a second portion, the first portion and the second portion differing in at least one of a strength, a hardness, a ductility, or a microstructure, wherein forming the first portion and the second portion comprises: receiving data from at least one heat sensor indicative of a temperature of a portion of the component; and modifying an amount of energy delivered by the first energy delivery device or a second energy device based at least partially on the received data from the at least one heat sensor to form the first portion and the second portion of the component body in-situ.

Claims

- 1.** An additive manufacturing system comprising: a first energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a second energy delivery device configured to deliver energy to the build surface of the component; a stage configured to support an additively-manufactured component; at least one heat sensor configured to capture data indicative of a temperature of a portion of a component; and one or more computing devices configured to: receive data from the at least one heat sensor; and control the first or the second energy device based at least partially on the received data from the at least one heat sensor to provide functionally-graded characteristics to the additively-manufactured component, in-situ, through modification of an amount of thermal energy delivered by the first energy delivery device or the second energy delivery device.
- 2.** The additive manufacturing system of claim 1, further comprising the additively-manufactured functionally-graded component, wherein the additively-manufactured functionally-graded component comprises a first portion and a second portion, wherein the first portion is different from the second portion in at least one of a strength, a hardness, a microstructure or a ductility.
- 3.** The additive manufacturing system of claim 1, wherein the one or more computing device is further configured to: determine an overall heat flux based on the data from the at least one heat sensor; and control the first energy delivery device and the second energy delivery device based on the overall heat flux.
- 4.** The additive manufacturing system of claim 1, wherein: the first energy deliver device is coincident with a central longitudinal axis of a deposition head, and the second energy delivery device is not coincident with the central longitudinal axis of the deposition head.
- 5.** The additive manufacturing system of claim 4, wherein: the first energy delivery device comprises a laser, and the second energy delivery device comprises a laser, an induction heater, an infrared heater, a gas impingement device, or a microwave heater.
- 6.** The additive manufacturing system of claim 1, further comprising a third energy delivery device.
- 7.** The additive manufacturing device of claim 1 wherein, to control the first energy delivery device or the second energy delivery device to provide the functionally-graded characteristics to the additively-manufactured component, the computing device is configured to control the first energy delivery device by modifying at least one of a power, a travel speed, a spot size, or a power density of the first or the second energy delivery device.
- 8.** The additive manufacturing system of claim 1, wherein the second energy delivery device is configured to deliver energy to the build surface of the component simultaneously with the first energy delivery device delivering energy to the build surface of the component.
- 9.** The additive manufacturing system of claim 1, wherein the second energy delivery device is configured to deliver energy to the build surface of the component subsequent to the first energy deliver device delivering energy to the build surface of the component.
- 10.** The additive manufacturing system of claim 1, wherein the one or more computing devices are configured to: determine a solidification rate of material surrounding the melt pool based on data received from an optical system, and control the second energy delivery device to modify the determined solidification rate.
- 11.** The additive manufacturing system of claim 1, wherein the second energy delivery device is configured to deliver energy to a local area of the build surface, wherein the local area comprises about 20 percent or less of the surface area of the build surface.
- 12.** The additive manufacturing system of claim 1, wherein the second energy delivery device is configured to deliver energy to a global area of the build surface.
- 13.** The additive manufacturing system of claim 1, further comprising a plurality of mass sensors, each mass sensor associated with a portion of the additive manufacturing system, and wherein the one or more computing devices is further configured to: receive data from the plurality of mass

sensors; determine an overall mass flux based on the data from the plurality of mass sensors; and control the powder delivery device based on the overall mass flux to generate the additively-manufactured component with functional characteristics.

14. The additive manufacturing system of claim 13, wherein the plurality of mass sensors comprises a powder flow monitoring system comprising: an illumination device configured to illuminate at least some powder the powder stream between the powder delivery device and the build surface; and an imaging device configured to image the illuminated powder at an image plane that intersects an longitudinal axis of a deposition head, and wherein the one or more computing devices is configured to determine a mass flow rate of powder from the powder delivery device using data from the powder flow monitoring system.

15. The additive manufacturing system of claim 1, further comprising a cooling device, and wherein the one or more computing devices is configured to control the cooling device to remove thermal energy from the build surface.

16. The additive manufacturing system of claim 13, wherein the plurality of mass sensors comprise a topology sensor configured to measure a topology of material added to the melt pool, wherein the one or more computing devices is further configured to determine a mass of powder added to the melt pool based on the topology of the material added to the melt pool and a density of the powder.

17. The additive manufacturing system of claim 16, wherein the one or more computing devices is further configured to determine a capture efficiency by dividing the mass of powder added to the melt pool by the mass of powder leaving the powder delivery device or dividing a mass rate of powder added to the melt pool by a mass flow rate of powder leaving the powder delivery device.

18. The additive manufacturing system of claim 14, wherein the one or more computing devices is configured to determine an overall heat flux based on the data from the at least one heat sensor and a capture efficiency.

19. The additive manufacturing system of claim 1, wherein the computing device is configured to: determine a solidification rate of material surrounding the melt pool based on data received from an optical system., and control the second energy delivery device to modify the determined solidification rate.

20. A method comprising: receiving, by one or more computing devices, data from at least one heat sensor configured to measure a temperature of a portion of a component of an additive manufacturing system, wherein the additive manufacturing system comprises a powder delivery device configured to direct a powder stream toward a melt pool in a build surface of an additively-manufactured component mechanically supported by a stage, a first energy delivery device configured to deliver energy to the build surface of a component to form the melt pool, a second energy delivery device configured to deliver energy to the build surface of the component; and controlling, by the one or more computing devices, the first or the second energy device based at least partially on the received data from the at least one heat sensor to provide functionally-graded characteristics to the additively-manufactured component, in-situ, through modification of an amount of thermal energy delivered by the first energy delivery device or the second energy delivery device.
